Controlled Generation Study

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Signature Seed: 94304

Course: ITAI 1378 – Module 07 Lab

Part A – Controlled Generation Figures

Figure A1 – Guidance Sweep

Caption: Guidance sweep using the fixed prompt. Seed = 94304, Steps = 30, Guidance = 3, 7, and 12.

Figure A2 – Steps Sweep

Caption: Steps sweep using the fixed prompt. Seed = 94304, Guidance = 7, Steps = 10, 25, and 50.

Figure A3

Caption: Seed sweep using Seeds = 123, 4567, and 89012. Guidance = 7, Steps = 25.

Figure A4 – Negative Prompt

Caption: Comparison between generation without a negative prompt and generation using the negative prompt:

"*blurry, low quality, watermark, text, extra objects*."

Seed = 94304, Steps = 30, Guidance = 7.

Part B – Image-to-Image Editing

Figure B1 – Original Photo

Figure B2 – Edited Photo

Prompt: A tropical beach at golden sunset with dramatic colorful clouds, warm cinematic lighting, crystal clear turquoise water, photorealistic, highly detailed.

Strength: 0.6

Seed: 94304

Steps: 30

Guidance: 7

Figure C1 – Counting Failure

Prompt: A photorealistic image of exactly seven red apples on a white table, evenly spaced, top-down view.

Observation: The prompt requested exactly seven apples, but the model generated twelve apples instead, demonstrating the counting limitation of diffusion models.

Part D – Process Log

Prompt Version 1

A Japanese garden with a bridge.

Observation: This initial prompt produced a recognizable scene, but it lacked detail and atmosphere.

Prompt Version 2

A peaceful Japanese garden with a wooden bridge, cherry blossom trees, a koi pond, photorealistic.

Observation: Additional descriptive elements improved the realism of the garden and produced a more detailed composition.

Prompt Version 3

A peaceful Japanese garden with a wooden bridge, cherry blossom trees, a koi pond, soft morning light, photorealistic, highly detailed.

Observation: Adding lighting and the phrase “highly detailed” produced sharper textures, richer colors, and more realistic results. This prompt was used consistently throughout Part A so that only the experimental parameters changed.

Reflection

This lab helped me better understand how different settings in a diffusion model affect the images it generates. By keeping the same prompt throughout Part A, I was able to clearly see how changing one parameter at a time affected the results. In **Figure A1**, increasing the guidance scale made the model follow the prompt more closely. The image with guidance 3 looked a little softer, while guidance 12 produced a sharper image with more vivid colors.

The steps sweep also showed how the number of inference steps affects image quality. In **Figure A2**, the image generated with 10 steps had less detail than the images generated with 25 and 50 steps. However, I noticed that the difference between 25 and 50 steps was much smaller, which showed that using more steps does not always improve the image significantly.

The seed experiment helped me understand why keeping the same seed is important when you want to reproduce the same image. Even though I kept the same prompt, guidance, and steps, changing only the seed created different versions of the Japanese garden. As shown in **Figure A3**, each image had a different arrangement of the bridge, trees, and pond while still matching the prompt.

In **Figure A4**, adding a negative prompt slightly improved the quality of the image by reducing unwanted artifacts and making the final result look cleaner. For Part B, the img2img model transformed my beach photo into a tropical sunset scene while keeping the original beach layout (**Figure B2**). Finally, **Figure C1** showed a common limitation of diffusion models. Even though the prompt asked for exactly seven apples, the model generated twelve instead, showing that these models can have difficulty following exact counting instructions. Overall, this lab gave me a better understanding of how diffusion models work and how changing different parameters affects the final generated image.